

RUA-RouterUA



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instituto de
telecomunicações

Milestone 5: Poster, Demo and Promotional video

PEI 2025/2026

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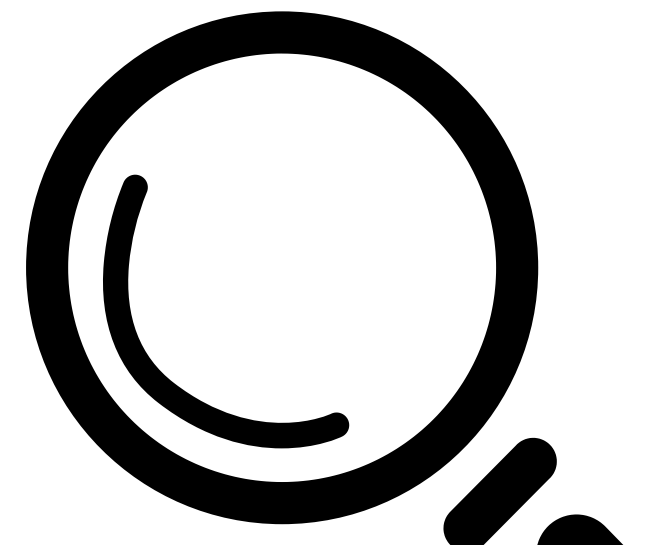
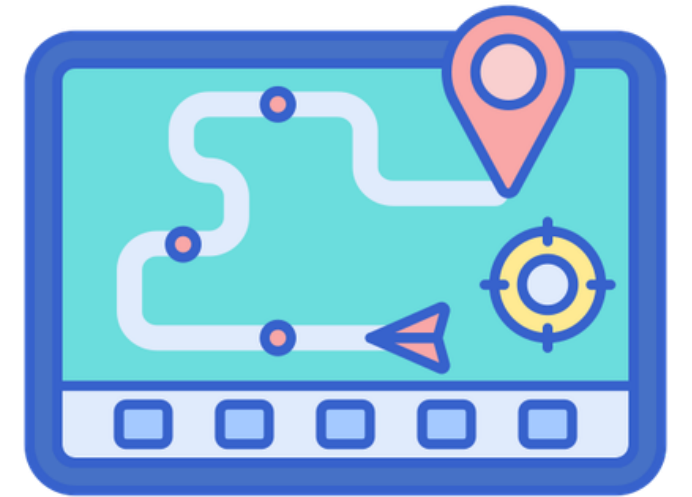
Abdelgader Mahmoud

Overview

- 01** Context and Goals
- 02** Poster
- 03** App Demo
- 04** Dashboard Demo
- 05** Promotional video

Context and Goals

- Improve real-time outdoor campus navigation for students, staff, and visitors
- Ensure safety with accessible and emergency evacuation routes
- Provide real-time congestion's maps and queues of UA



Calendar

**beginning of the
second semester**

M1

M2

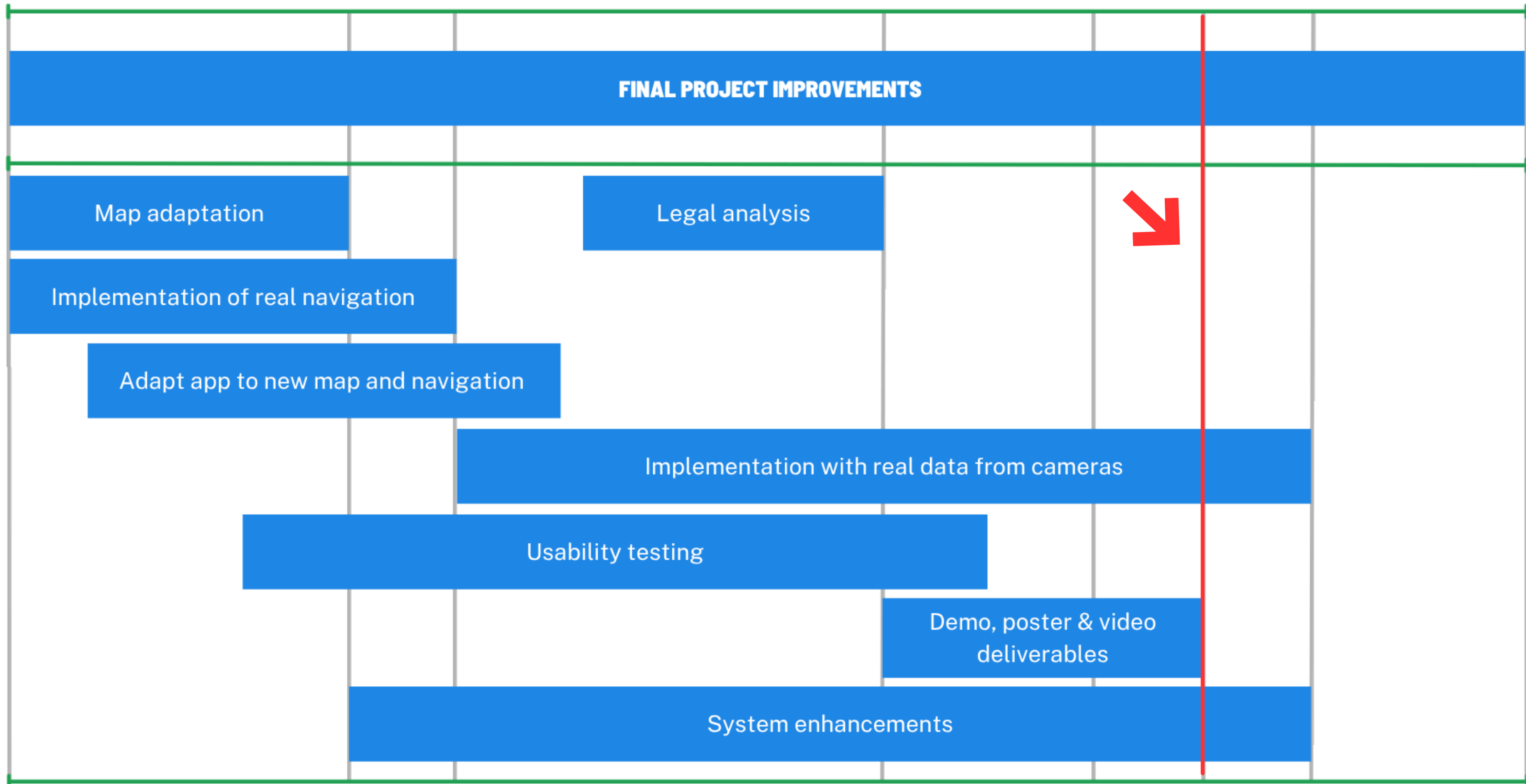
M3

M4

M5

M6

students@deti



Poster



RUA-RoutingUA

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Project in Informatic Engineer, 3rd year, LEI

2026



Abstract

Mobile outdoor navigation app for University of Aveiro campus, featuring adaptive event-based maps (external management interface) and privacy-friendly AI cameras for crowd counting and queues predictions . Targets students and event visitors. The system also includes emergency evacuation routes and step-free accessible navigation, with all location data anonymised and automatically expired.

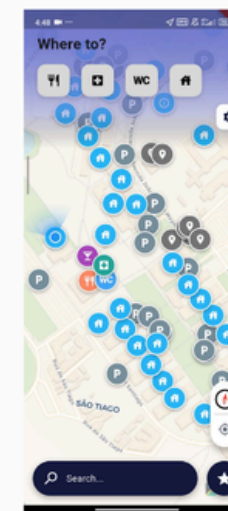


Fig. 1 - Main Users interface

Workflow

The user opens the app and gives one-time consent for anonymous GPS. They can choose categories such as “Canteens”, and the map then shows all canteens with current waiting time and estimated travel time to each. The user selects their preferred option, and the app calculates the best route. If the accessibility filter is toggled on, only step-free accessible routes are offered. During navigation, if the heatmap option is active, live crowd density overlays appear on the map. While en route, if a queue becomes significantly long, the system proactively suggests a reroute to a nearby canteen with a shorter wait.

Platforms and functionalities

- Mobile PWA - Outdoor turn-by-turn navigation with live congestion heatmaps and an accessibility mode for step-free routes.
- Routing service - A* pathfinding service that adapts in real time based on crowd density and wait-time estimates.
- Congestion & Wait-Time Service - Ingests anonymous GPS grids and camera counts, and computes queue predictions.
- Event Map Manager - External interface for organisers to upload custom maps, temporary POIs, and paths for fairs, conferences and campus events.

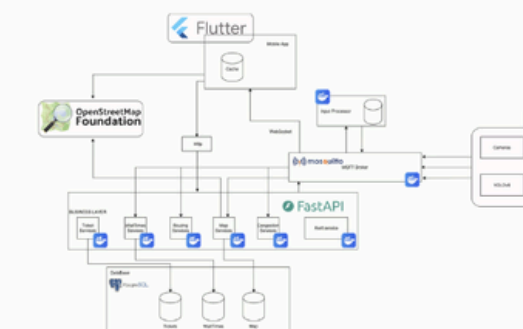


Fig. 2 - System Architecture

Summary

The current prototype demonstrates the technical viability of privacy-first real-time campus navigation using both live and pre-recorded data on the University of Aveiro campus. Live A* routing reacts dynamically to crowd-based edge weights, anonymous camera counting via ONNX outputs (only occupancy grids), and queue predictions update every few seconds. Accessible routes are fully separated by the nodes defined in the map service. The containerised application is ready for real-world pilot deployment.

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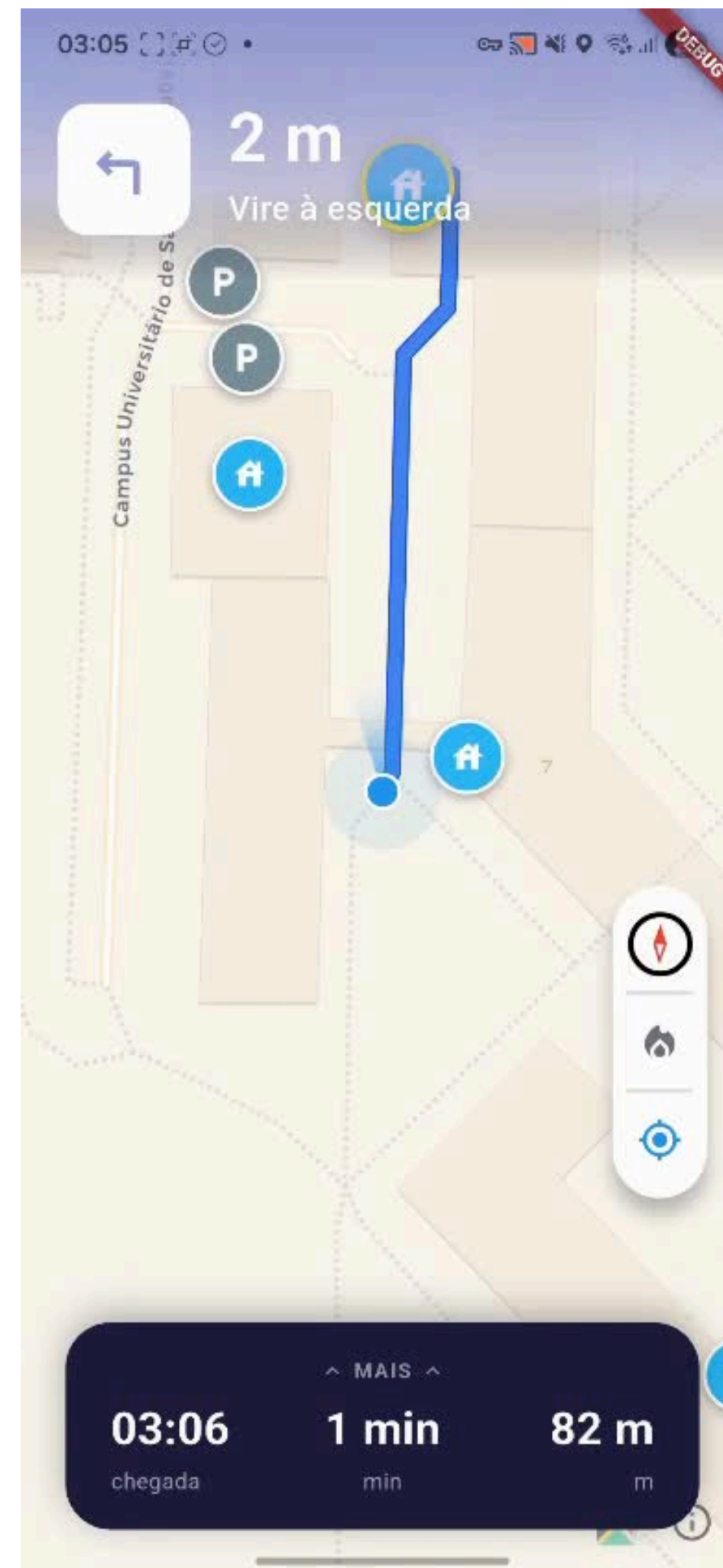
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deti departamento de electrónica,
telecomunicações e informática

App Demo

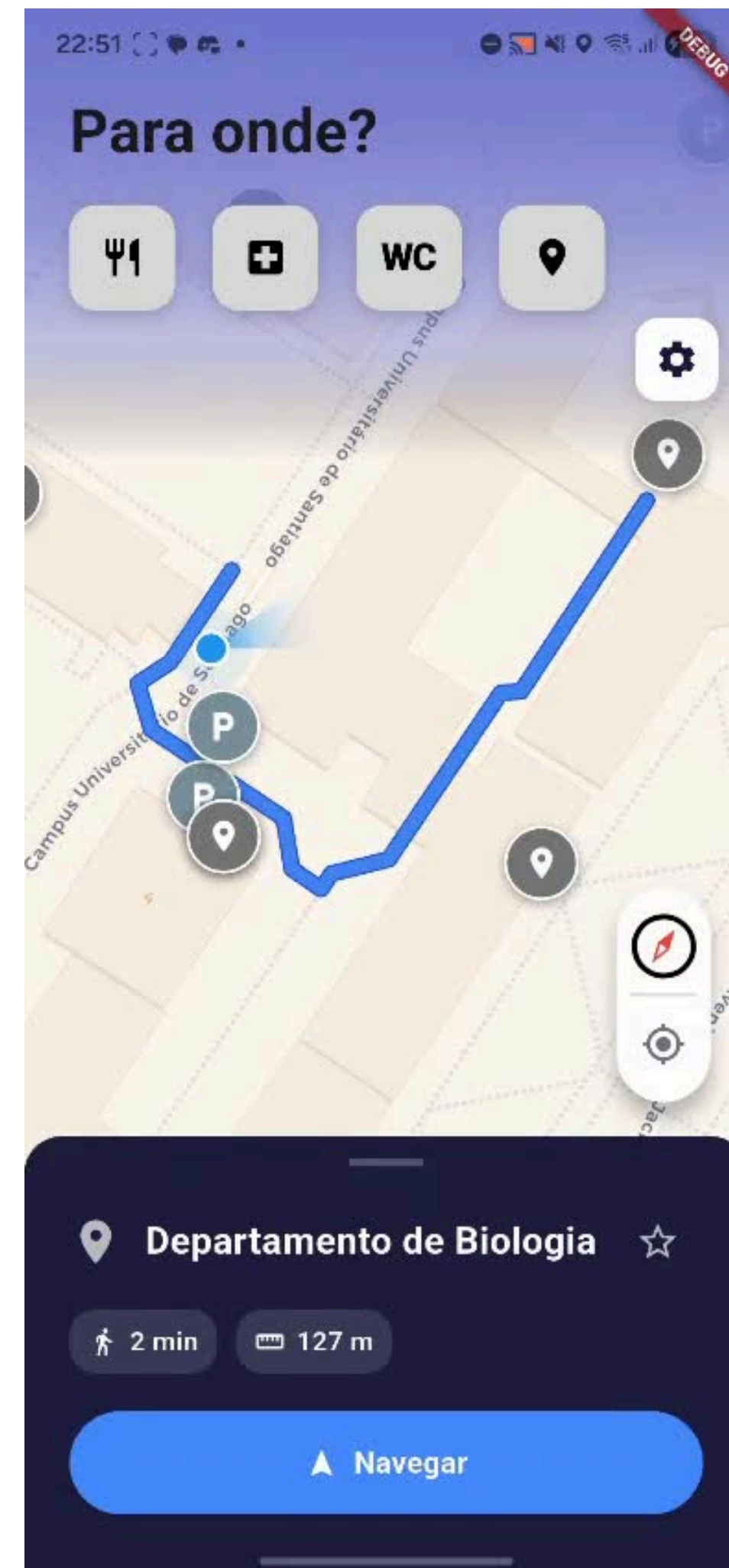
Demo

Simple navigation



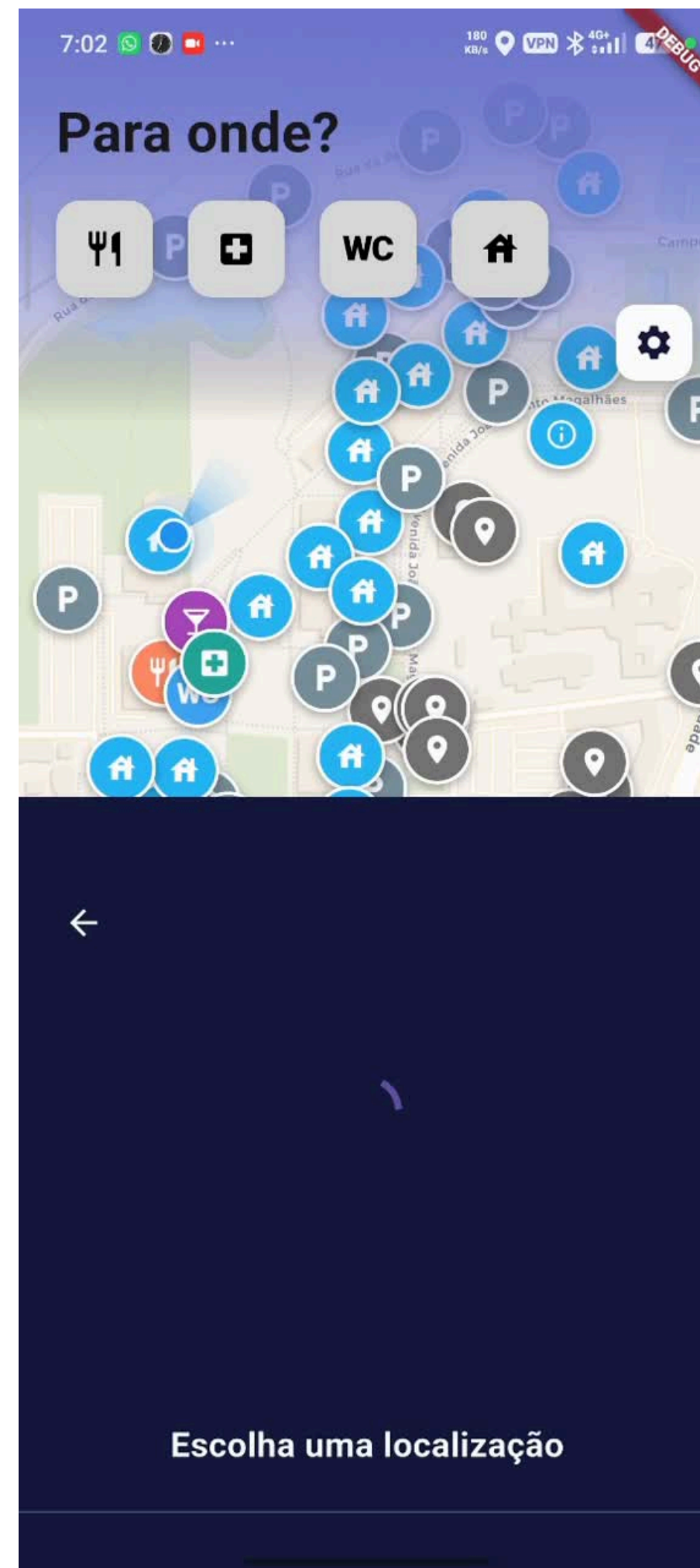
Demo

Navigation with Re-
routing



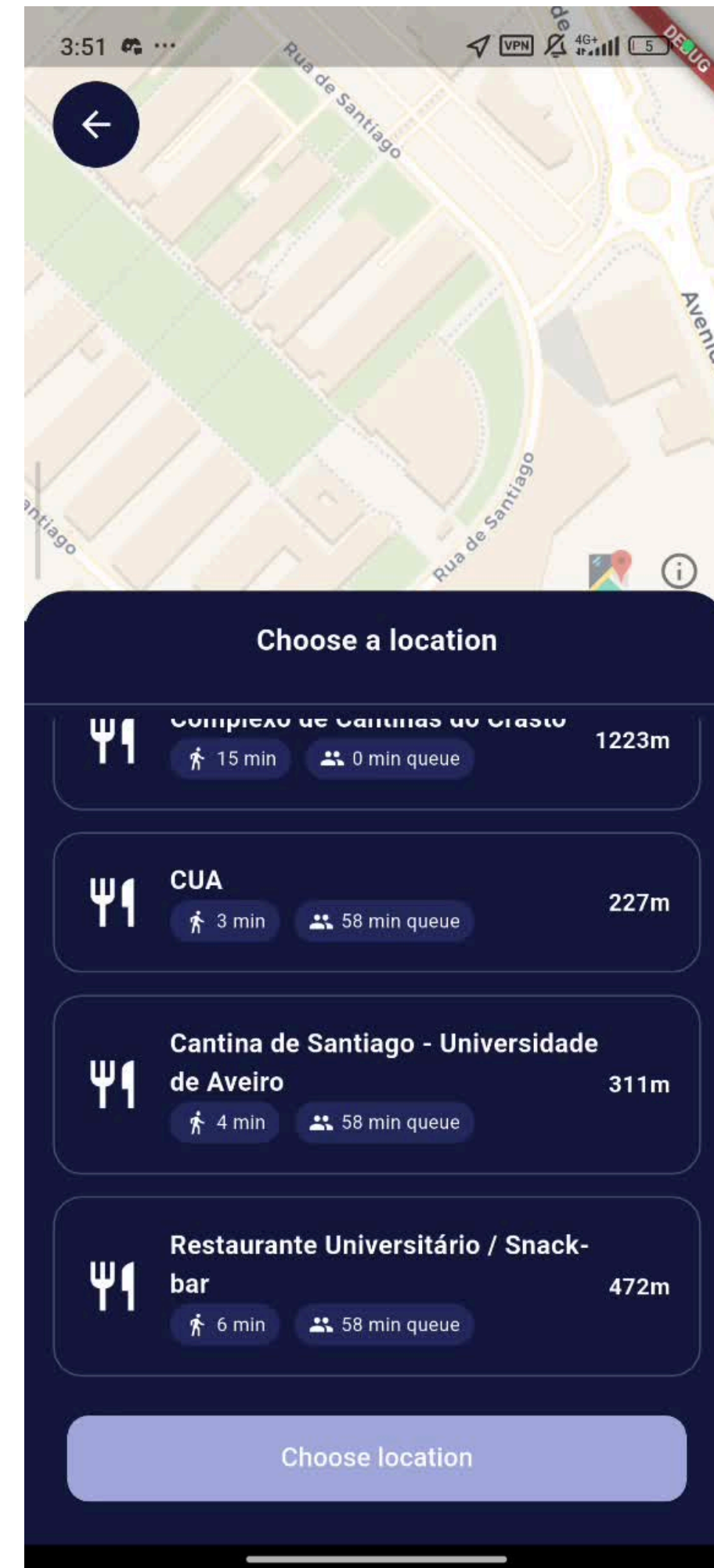
Demo

Filtering



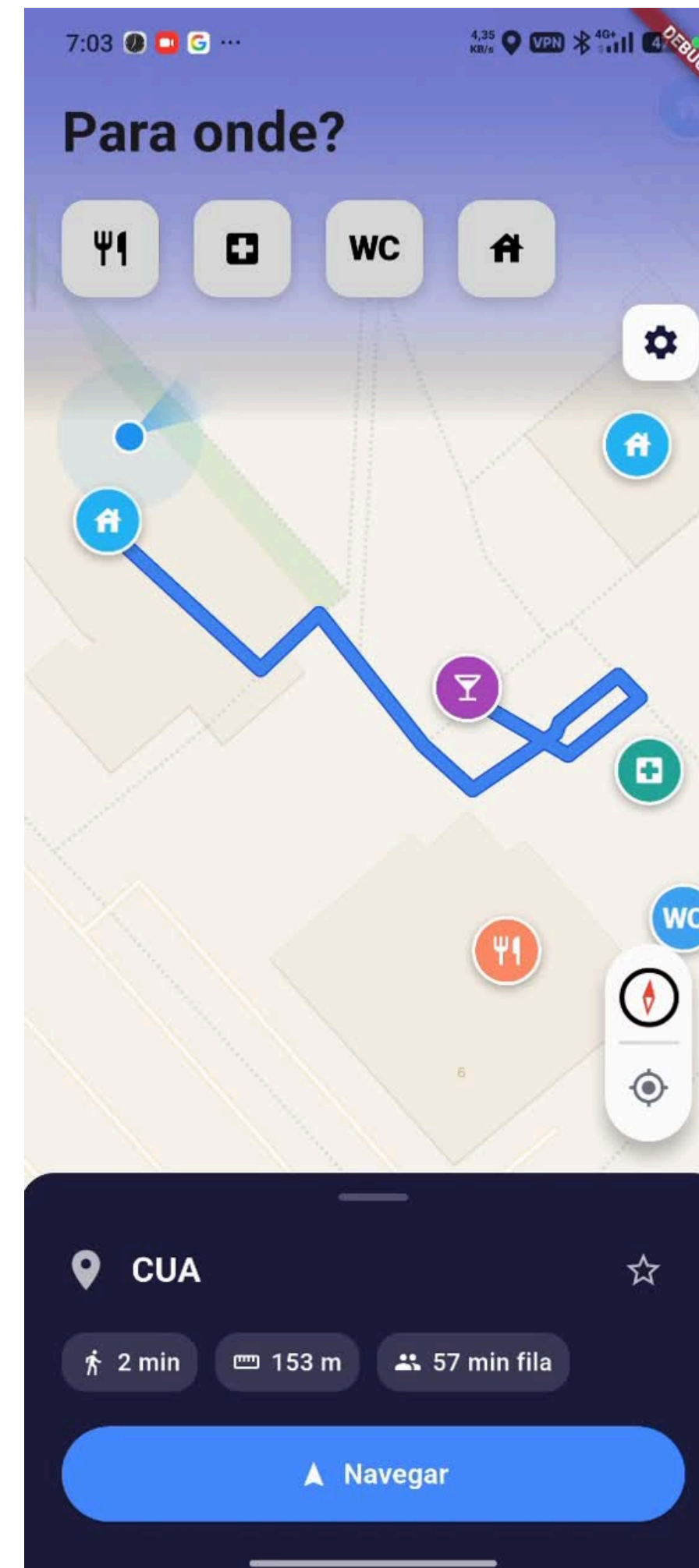
Demo

Wait-times



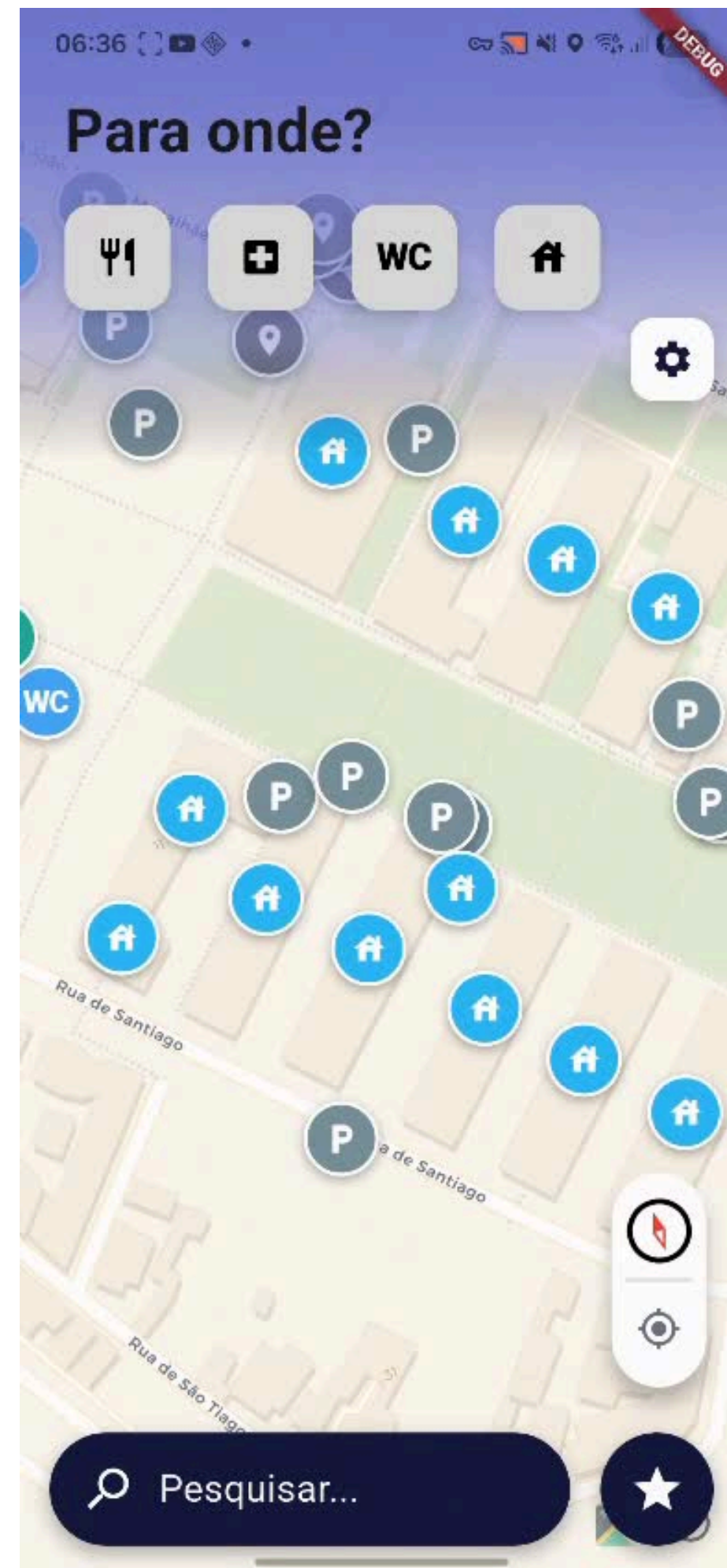
Demo

Accessibility switch



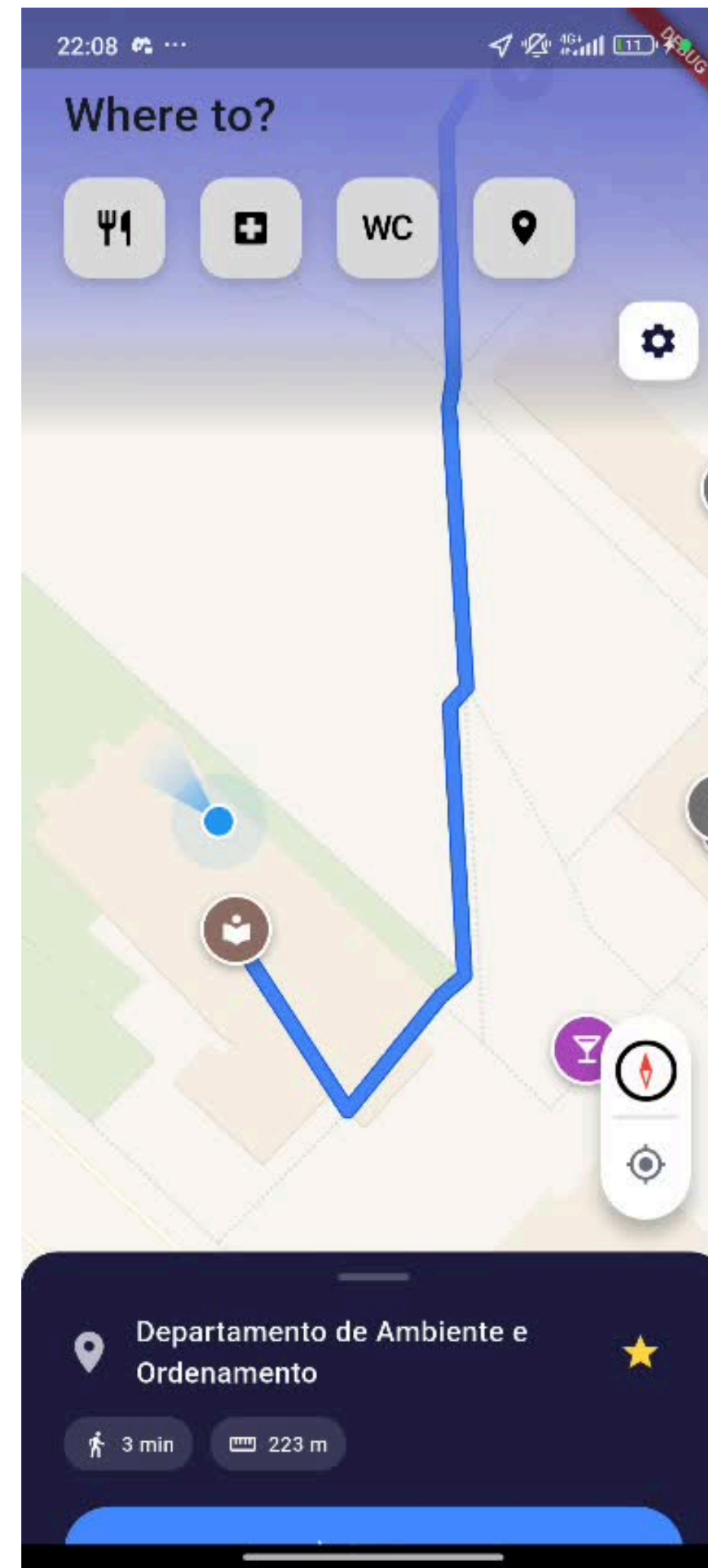
Demo

Heat-map



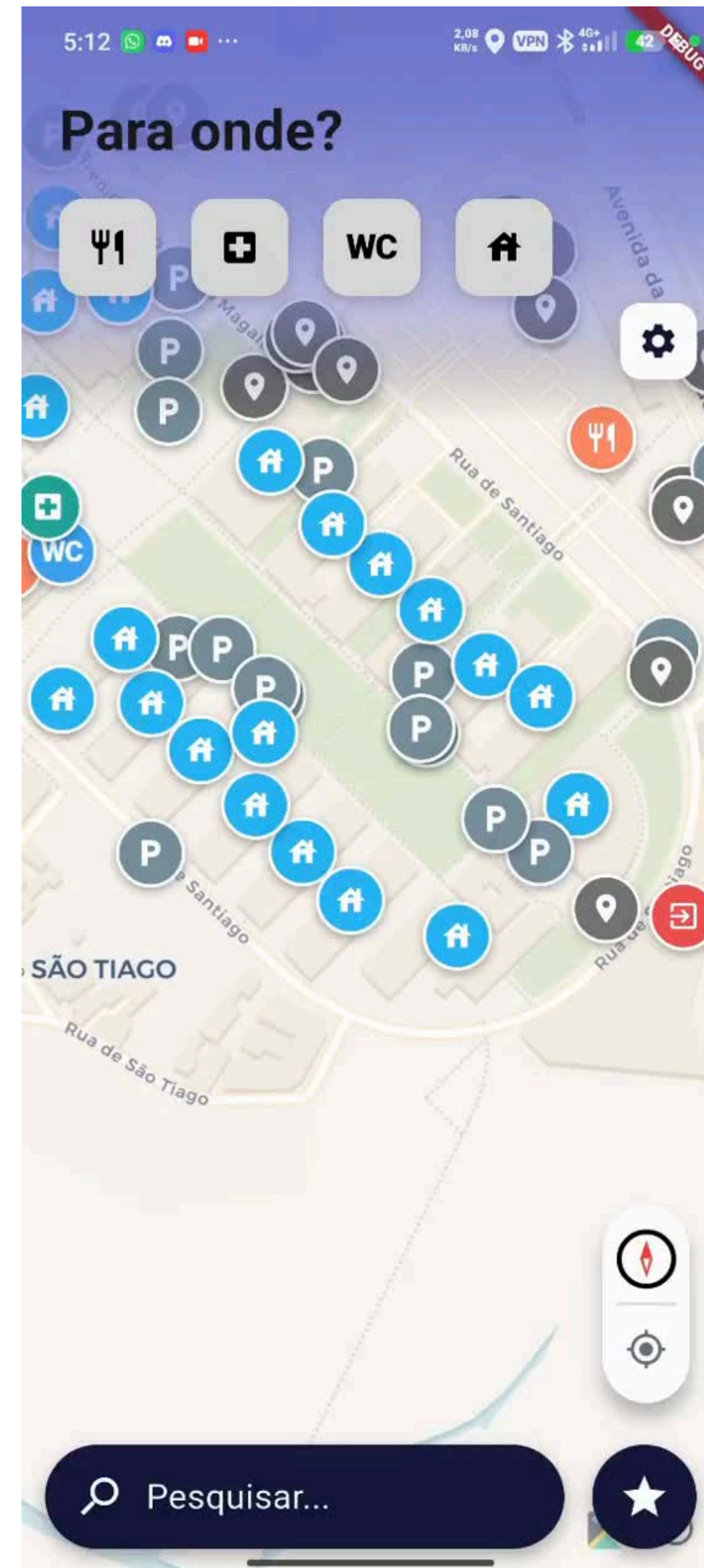
Demo

Favourites



Demo

Alert



Dashboard Demo



niversity of Aveiro

Promotional video



Questions?



Thank You!

Project Microsite:

